Definition-Only RAG for Financial QA: Average Gains and a Diagnostic Boundary

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Introduction

► Answering a specialized financial question often turns on knowing what a particular term means, and a model that has not internalised the term can fail for that reason alone. Qwen-2.5-7B answers *American Depositary* Receipt where the answer is Guilder Share; one sentence of Investopedia flips it.

► Scale has not removed the problem: on MMLU finance subjects a frontier model still misses one question in eight, and 7–8B models miss **40–50%**.

► Retrieval-augmented generation is the standard remedy, and a short glossary definition is the smallest and most auditable unit of evidence it can supply against that failure.

► Prior work reports that a model's ability to define domain terms correlates with its performance in that domain (Jhirad et al., 2023), which raises the question of whether the same measurement can also decide when to retrieve.

RQ1 · Input Level

Does definition-only retrieval improve financial QA on average?

RQ2 · Routing Level

Does the model's own term knowledge predict when retrieval helps?

Related work

Definition modeling

probes what a model has internalized about a domain.

- Noraset et al. 2017; Gardner et al. 2022
- Finance: Jhirad et al. 2023. Multilingual: Marrese-Taylor et al. 2025

Selective and adaptive RAG

assumes some observable property of the question or model state carries enough signal to route between retrieve and no-retrieve.

- Lewis et al. 2020; Asai et al. 2024; Jeong et al. 2024

Uncertainty-style routing baselines

Semantic entropy (Farquhar et al. 2024) and option-margin signals.

- Wang et al. 2024: first-token MCQ scoring can disagree with the model's final text answer.

We test the **narrowest version** of that assumption in finance, using definition-level term knowledge, the most semantically transparent candidate signal this domain offers.

Method and Data

Diagnostic. The model writes the definition itself. Measures what it already holds, no retrieval.

Interventional. We supply gold definitions through retrieval and measure what changes downstream.

Model panel. Qwen-2.5-7B and 14B, Llama-3.1-8B, Mistral-Small-24B, GPT-5.2: four families from 7B to a frontier closed model, so a capability-dependent effect is separable from a universal one.

Retrieval pipeline. A lexical retriever and a dense sentence-encoder each propose candidates from a fixed glossary of **13,609** Investopedia terms, the two ranked lists are fused, a cross-encoder reranks them, and the **top-3** snippets are injected above the question stem. Option letters are scored by log-probability at temperature 0, fixed across every experiment.

Paired evaluation. Every question is answered twice, with the definitions and without, so each lands in one of four states: correct in both, **fixed**, **broken**, or wrong in both. Net gain is the fix rate minus the break rate, and an oracle that retrieved only where it helps would collect the fixes without paying the breaks.

Resource	Size	Role
Investopedia glossary	13,609	index + reference
MMLU Finance	1,124 q	primary benchmark
XFinBench TU	1,790 q	replication
Term annotation	995 q	per-question map
Invented terms	50	memorisation control

Table 1. The glossary, the two benchmarks, and the resources derived from them.

Contributions. We release a **human-validated term annotation** mapping each MMLU Finance question to the terms its answer depends on: two LLM annotators propose (κ = 0.76), 2,233 pairs adjudicated by hand, 995 term-bearing questions. No such mapping existed, and it is what connects term knowledge to per-question outcomes.

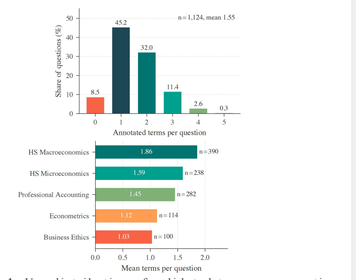


Figure 1. Up: distribution of validated terms per question. Down: per-subject term distribution across the five MMLU Finance sub-subjects.

Results

Definition-only retrieval helps on average, and the gain is confined to a mid-capability band on **both** benchmarks.

Model	Base	+RAG	Δ
MMLU Finance (n = 1,124)			
Qwen-2.5-7B	60.8	64.1	+3.3
Qwen-2.5-14B	68.2	68.1	−0.1
Llama-3.1-8B	52.1	57.6	+5.4
Mistral-S-24B	72.3	72.5	+0.2
GPT-5.2	87.8	88.1	+0.3
XFinBench TU (n = 1,790)			
Qwen-2.5-7B	70.5	75.3	+4.8
Qwen-2.5-14B	77.7	76.8	−0.8
Llama-3.1-8B	61.5	63.5	+2.0
Mistral-S-24B	78.9	78.9	±0.0
GPT-5.2	87.6	88.5	+0.9

Table 2. Accuracy with and without definition-only RAG. Coloured Δ: 95% CI excludes zero (paired bootstrap, 2,000 resamples).

The split follows neither size nor family. No stalling model gains even in its own weakest term-knowledge quartile.

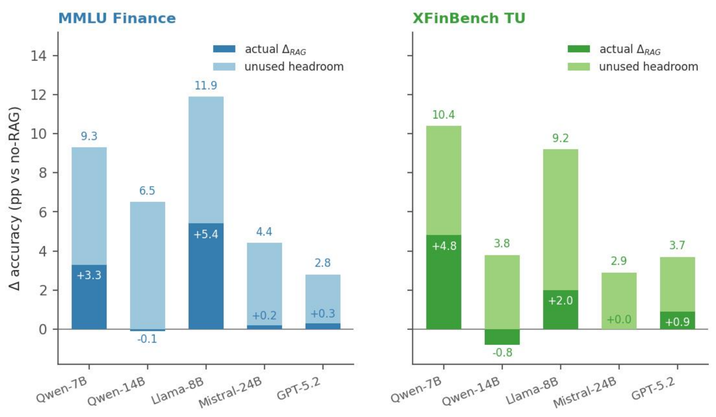


Figure 2. Actual ΔRAG vs implementable-oracle upper bound.

The gain is the definition, not leakage. Only 20 of 3,228 injected snippets contain a full answer string, and removing them changes nothing.

Condition	Acc.	vs RAG
No RAG (baseline)	60.8	−2.9
RAG (as reported)	63.7	—
drop answer-matching	63.7	±0.0
mask all term names	61.9	−1.8
Invented terms (n = 50, unmemorisable)		
GPT-5.2	36.0 →	100.0
Qwen-2.5-7B	30.0 →	100.0

Table 3. Controls on the source of the gain.

A positive control. On unmemorisable terms the frontier model uses a definition as completely as the mid-capability one, so its flat Δ is not an inability to read one.

Analysis: Routing Retrieval on Term Knowledge

A net of zero hides two-sided movement. Qwen-2.5-14B sits at Δ ≈ 0 while retrieval still changes 147 of 1,124 answers.

Term knowledge does not predict per question. All 40 cells lie inside |r| < 0.06 (permutation p = 0.91).

No cheap internal signal approaches the oracle.

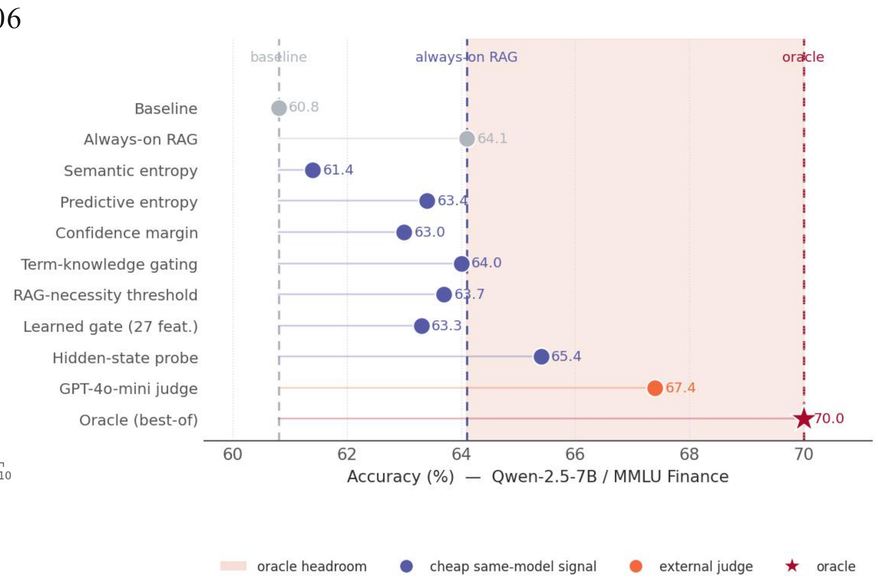


Figure 5. Routing-signal accuracies against the oracle gap.

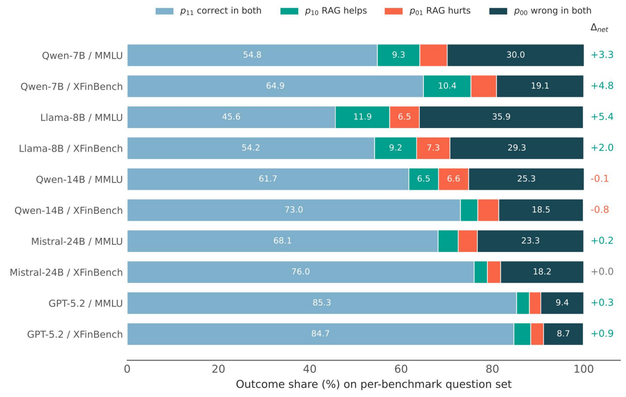


Figure 3. Stacked-bar visualisation of the paired RAG-removal decomposition.

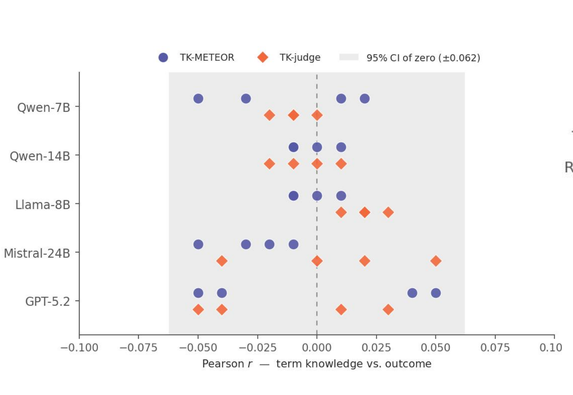


Figure 4. Per-question correlations as a forest plot.

What the three panels say. Only the external judge beats always-on retrieval, at 67.4 against 64.1 with the oracle at 70.0, and that judge is itself a second and stronger model. Filtering which of the retrieved definitions to inject, rather than whether to retrieve at all, does not help either. The injected text changes the scores the model assigns to the four options, and whether that helps or hurts depends on the question, the option set and the snippet together.

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github.com/youngandlei/definition-only-rag